

The Exponential Hinge

One mathematical form, three different physical causes

IB Physics A3 worksheet

Central idea. In each case below, a quantity decreases exponentially:

$$N = N_0 e^{-\lambda t}, \quad Q = Q_0 e^{-t/RC}, \quad I = I_0 e^{-\mu x}.$$

The mathematics is similar, but the physical meaning of the variable and constant changes from case to case.

In each panel:

1. Identify the quantity that is decreasing.
2. Identify the independent variable.
3. Identify the constant that sets the scale of the decrease.
4. State what physical model gives that constant its value.
5. Compare the meaning of “half” in the three cases.

1. Radioactive decay

Core relations

$$N = N_0 e^{-\lambda t}, \quad A = \lambda N, \quad t_{1/2} = \frac{\ln 2}{\lambda}.$$

A sample begins with

$$N_0 = 8.0 \times 10^6$$

unstable nuclei. Its decay constant is

$$\lambda = 2.0 \times 10^{-4} \text{ s}^{-1}.$$

Tasks

1. Write an expression for N after time t .
2. Calculate N after 2000 s.
3. Find the half-life.
4. State what determines λ .
5. State clearly the difference between N and A .

6. Complete the sentence: The exponential form tells me _____.

Explore

- What is actually decreasing?
- Why can the behaviour of a large sample be regular even though individual decay is random?
- What are the units of λ , and why?

2. Capacitor discharge

Core relations

$$Q = Q_0 e^{-t/RC}, \quad V = V_0 e^{-t/RC}, \quad I = I_0 e^{-t/RC}.$$

A capacitor of capacitance

$$C = 220 \mu\text{F}$$

discharges through a resistor of resistance

$$R = 4.7 \text{ k}\Omega.$$

Tasks

1. Calculate the time constant RC .
2. Write an expression for the charge Q after time t .
3. Find the fraction of the original charge remaining after one time constant.
4. Find the time for the charge to fall to half its initial value.
5. State what physical features determine the rate of discharge.
6. Which other quantities share the same exponential form during discharge?

Explore

- What does the time constant mean physically?
- Why is the half-value time not equal to RC ?
- What are the units of RC , and why?

3. Gamma attenuation

Core relations

$$I = I_0 e^{-\mu x}, \quad x_{1/2} = \frac{\ln 2}{\mu}.$$

A beam of gamma radiation passes through lead. The attenuation coefficient is

$$\mu = 1.5 \text{ cm}^{-1}.$$

Tasks

1. Write an expression for the transmitted intensity after thickness x .
2. Find the intensity after 2.0 cm, as a fraction of I_0 .
3. Find the half-value thickness.
4. State what determines μ .
5. Explain why this is exponential in *distance* rather than in time.
6. State the units of μ , and explain why they make sense.

Explore

- What is decreasing here?
- Why does a thicker absorber give a smaller transmitted intensity?
- What is the difference between half-value thickness and one attenuation length $1/\mu$?

Synthesis

Complete the table.

Case	Decreasing quantity	Independent variable	Characteristic constant	What determines the constant?
Radioactive decay				
Capacitor discharge				
Gamma attenuation				

Final questions

1. What is mathematically similar across all three cases?
2. What is physically different across all three cases?
3. Why do students often confuse half-life, time constant, and half-value thickness?
4. Complete this sentence:
The exponential form gives the shape of the change; the physics gives _____.

The Exponential Hinge

Teacher key

Purpose. Students compare three exponential relations and identify:

- the quantity changing;
- the independent variable;
- the characteristic constant;
- the physical model that determines that constant.

Main learning point. The exponential relation is the shared hinge. The physical interpretation of the constant comes from nuclear physics, circuit behaviour, or radiation–matter interaction.

1. Radioactive decay

Model answers

1.
$$N = 8.0 \times 10^6 e^{-2.0 \times 10^{-4} t}.$$
2. At $t = 2000$ s,
$$N = N_0 e^{-0.4} = 8.0 \times 10^6 (0.6703) \approx 5.36 \times 10^6.$$
3.
$$t_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{2.0 \times 10^{-4}} \approx 3.47 \times 10^3 \text{ s}.$$
4. λ is set by the nuclear instability of the isotope.
5. N is the number of undecayed nuclei. A is the activity, the rate of decay:
$$A = \lambda N.$$

Useful remarks

- λ has units s^{-1} .
- Large-sample regularity can be discussed qualitatively.
- Students often confuse N and A .

2. Capacitor discharge

Model answers

1.
$$RC = (4.7 \times 10^3)(220 \times 10^{-6}) = 1.034 \text{ s} \approx 1.03 \text{ s}.$$
2.
$$Q = Q_0 e^{-t/1.03}.$$
3. After one time constant,
$$Q = Q_0 e^{-1} \approx 0.368 Q_0.$$
4. Half-value time:
$$\frac{Q}{Q_0} = \frac{1}{2} \quad \Rightarrow \quad t = RC \ln 2 = (1.034)(0.693) \approx 0.716 \text{ s}.$$
5. The rate is determined by the resistance R and capacitance C .
6. V and I also vary exponentially during discharge.

Useful remarks

- RC is the time constant.
- Students often think “one time constant” means “half”.
- RC has units of seconds.

3. Gamma attenuation

Model answers

1.
$$I = I_0 e^{-1.5x}$$
if x is measured in cm.
2. At $x = 2.0$ cm,
$$\frac{I}{I_0} = e^{-3.0} \approx 0.0498.$$
3.
$$x_{1/2} = \frac{\ln 2}{\mu} = \frac{0.693}{1.5} \approx 0.462 \text{ cm}.$$
4. μ depends on the material and the radiation interaction.
5. The beam loses intensity as it passes through material, so the independent variable is thickness x , not time.
6. μ has units of inverse length, here cm^{-1} .

Useful remarks

- Keep the terminology “half-value thickness” clear.
- $1/\mu$ is the attenuation length, not the half-value thickness.

Synthesis answers

Case	Decreasing quantity	Independent variable	Characteristic constant	What determines it?
Radioactive decay	N	t	λ	nuclear instability
Capacitor discharge	Q, V, I	t	RC	resistance and capacitance
Gamma attenuation	I	x	μ	absorber material and radiation interaction

Model statements

- The mathematical similarity is exponential decrease from an initial value.
- The physical difference lies in what is changing, what the independent variable is, and what determines the constant.
- Half-life, time constant, and half-value thickness are often confused because all three describe a scale of exponential change, but they are not the same quantity.
- Good completion sentence:
The exponential form gives the shape of the change; the physics gives the meaning and value of the constant.